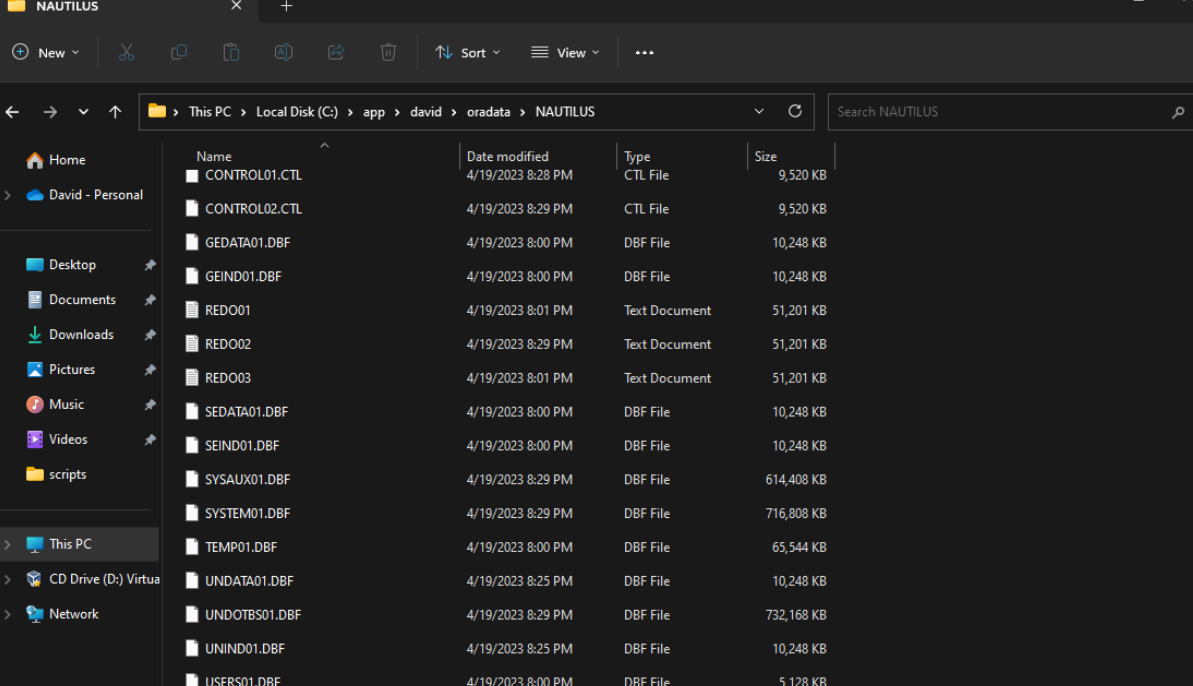
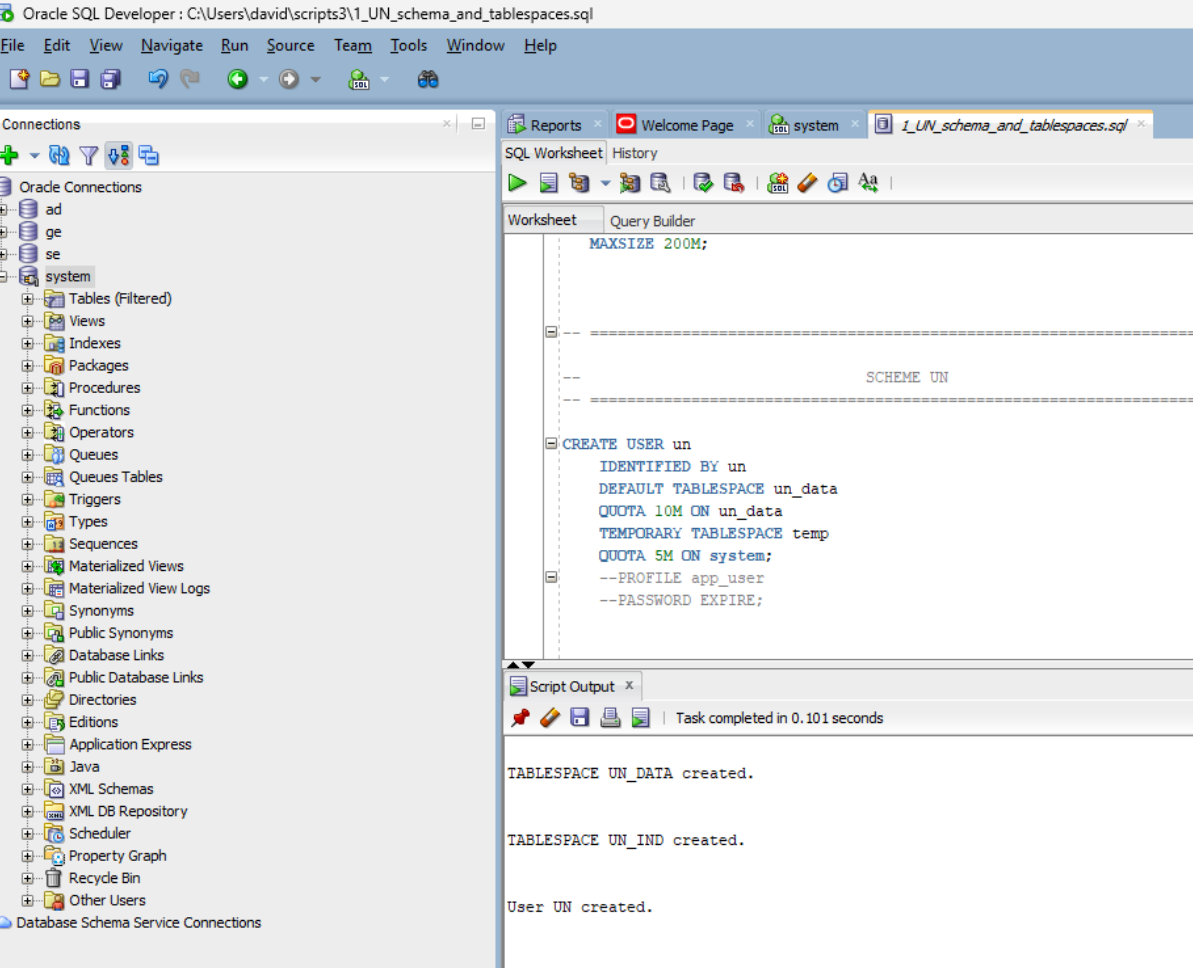


Evidence Databases

1.UN Schema Creation



Name	Date modified	Type	Size
CONTROL01.CTL	4/19/2023 8:28 PM	CTL File	9,520 KB
CONTROL02.CTL	4/19/2023 8:29 PM	CTL File	9,520 KB
GEDATA01.DBF	4/19/2023 8:00 PM	DBF File	10,248 KB
GEIND01.DBF	4/19/2023 8:00 PM	DBF File	10,248 KB
REDO01	4/19/2023 8:01 PM	Text Document	51,201 KB
REDO02	4/19/2023 8:29 PM	Text Document	51,201 KB
REDO03	4/19/2023 8:01 PM	Text Document	51,201 KB
SEDATA01.DBF	4/19/2023 8:00 PM	DBF File	10,248 KB
SEIND01.DBF	4/19/2023 8:00 PM	DBF File	10,248 KB
SYS_AUX01.DBF	4/19/2023 8:29 PM	DBF File	614,408 KB
SYSTEM01.DBF	4/19/2023 8:29 PM	DBF File	716,808 KB
TEMP01.DBF	4/19/2023 8:00 PM	DBF File	65,544 KB
UNDATA01.DBF	4/19/2023 8:25 PM	DBF File	10,248 KB
UNDOTBS01.DBF	4/19/2023 8:29 PM	DBF File	732,168 KB
UNIND01.DBF	4/19/2023 8:25 PM	DBF File	10,248 KB
USERS01.DBF	4/19/2023 8:00 PM	DBF File	5,128 KB



Oracle SQL Developer : C:\Users\david\scripts3\1_UN_schema_and_tablespaces.sql

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Connections

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SQL Worksheet History

Worksheet Query Builder

```
MAXSIZE 200M;
```

SCHEME UN

```
CREATE USER un
  IDENTIFIED BY un
  DEFAULT TABLESPACE un_data
  QUOTA 10M ON un_data
  TEMPORARY TABLESPACE temp
  QUOTA 5M ON system;
--PROFILE app_user
--PASSWORD EXPIRE;
```

Script Output x

Task completed in 0.101 seconds

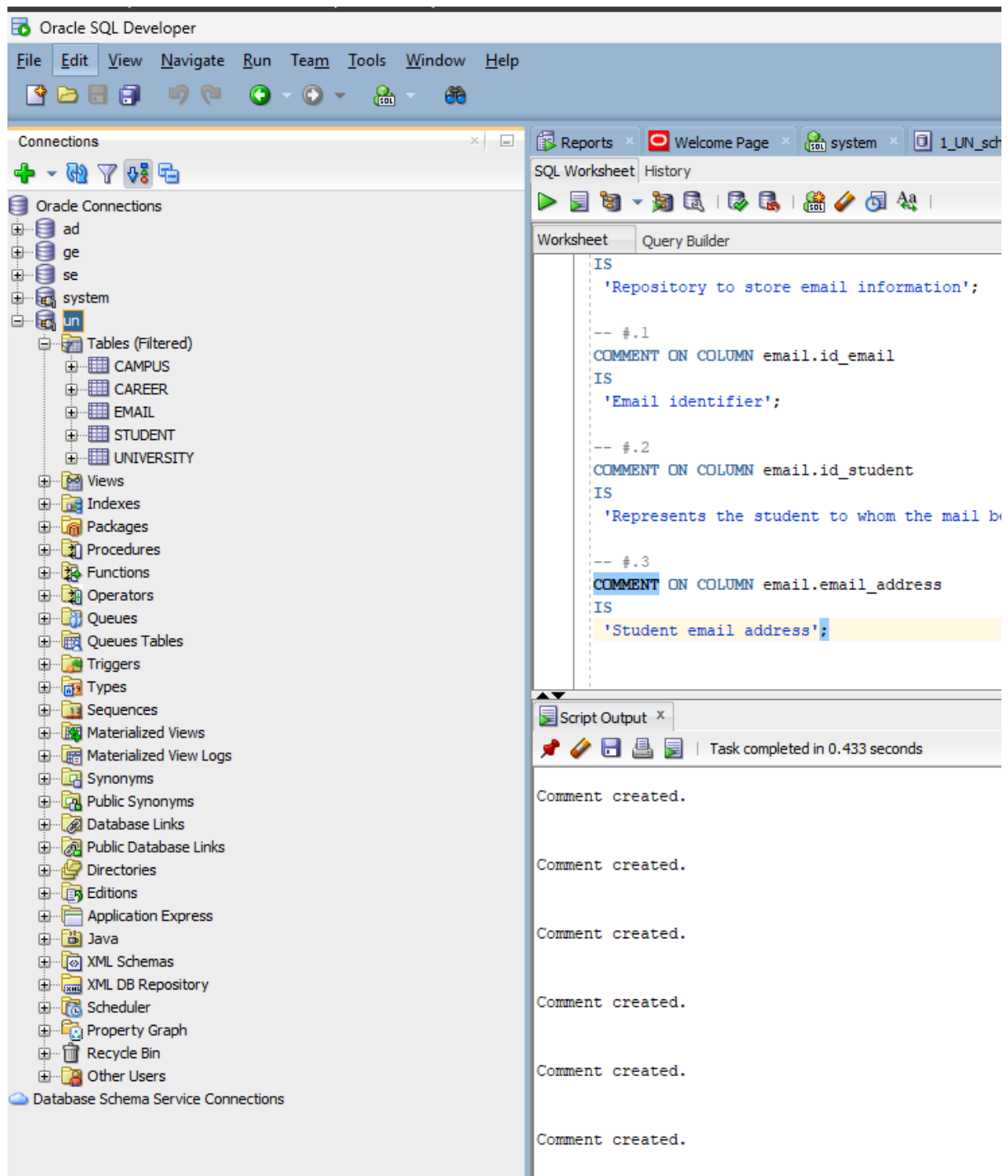
TABLESPACE UN_DATA created.

TABLESPACE UN_IND created.

User UN created.

Grant succeeded.

2.Model y tables



*Note: the model is attached in the folder.

3. Creation of primary and foreign keys (using sequences).

Sequences:

12	SE_CAMPUS
12	SE_CAREER
12	SE_EMAIL
12	SE_STUDENT
12	SE_UNIVERSITY

PK UNIVERSITY:

Columns
Data
Model
Constraints
Grants
Statistics
Triggers

Actions...

	CONSTRAINT_NAME	CONSTRAINT_TYPE	STATUS
1	PK_UNIVERSITY	Primary_Key	(

PK and FK CAMPUS:

CONSTRAINT_NAME	CONSTRAINT_TYPE
1 FK_CAMPUS_UNIVERSITY	Foreign_Key
2 PK_CAMPUS	Primary_Key

PK and FK CAREER:

Actions...	
CONSTRAINT_NAME	CONSTRAINT_TYPE
1 FK_CAREER_CAMPUS	Foreign_Key
2 PK_CAREER	Primary_Key

PK and FK STUDENT:

			Actions...
--	--	--	------------

Columns	CONSTRAINT_NAME	CONSTRAINT_TYPE
1	FK_STUDENT_CAREER	Foreign_Key
2	PK_STUDENT	Primary_Key

PK and FK EMAIL:

Columns	Data	Model	Constraints	Grants	Statistics	Triggers
			▼ Actions...			
		CONSTRAINT_NAME		CONSTRAINT_TYPE		
1		FK_EMAIL_STUDENT		Foreign_Key		
2		PK_EMAIL		Primary_Key		

4. Check constraints

UNIVERSITY Constraints:

CONSTRAINT_NAME	CONSTRAINT_TYPE	STATUS
1 PK_UNIVERSITY	Primary_Key	(
2 UNIVERSITY_UNIVERSITYNAME_NN	Check	(

CAMPUS Constraints:

CONSTRAINT_NAME	CONSTRAINT_TYPE
CAMPUS_CAMPUSNAME_NN	Check
CAMPUS_IDUNIVERSITY_NN	Check

CAREER Constraints:

Columns	Data	Model	Constraints	Grants	Statistics	Triggers
Actions...						
	CONSTRAINT_NAME	CONSTRAINT_TYPE				
1	CAREER_CAREERNAME_NN	Check				
2	CAREER_IDCAMPUS_NN	Check				
3	FK_CAREER_CAMPUS	Foreign_Key				
4	PK_CAREER	Primary_Key				

STUDENT Constraints:

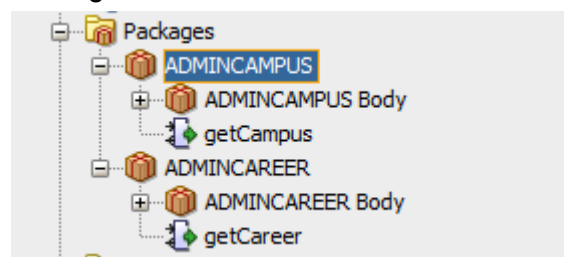
CONSTRAINT_NAME	CONSTRAINT_TYPE					
FK_STUDENT_CAREER	Foreign_Key					
PK_STUDENT	Primary_Key					
STUDENT_FIRSTLASTNAME_NN	Check					
STUDENT_FIRSTNAME_NN	Check					
STUDENT_IDCAREER_NN	Check					
STUDENT_SECONDLASTNAME_NN	Check					
STUDENT_STUDENTCARD_NN	Check					
STUDENT_VALIDSTUDENTCARD_MIN	Check					

EMAIL Constraints:

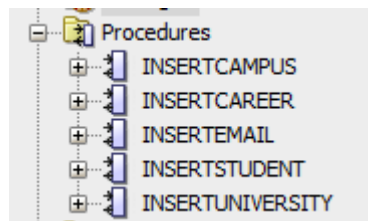
Columns	Data	Model	Constraints	Grants	Statistics	Triggers
Actions...						
	CONSTRAINT_NAME	CONSTRAINT_TYPE				
1	EMAIL_EMAILADDRESS_NN	Check				
2	EMAIL_IDSTUDENT_NN	Check				
3	FK_EMAIL_STUDENT	Foreign_Key				
4	PK_EMAIL	Primary_Key				

5.Packages, procedures, functions

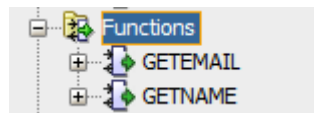
Packages:



Procedures:



Functions:



Tables:

Table university

	ID_UNIVERSITY	UNIVERSITY_NAME
1	0	Technological of Costa Rica

Table campus

	ID_CAMPUS	ID_UNIVERSITY	CAMPUS_NAME
1	0	0	San Jose
2	1	0	Cartago
3	2	0	Limon

Table career

	ID_CAREER	ID_CAMPUS	CAREER_NAME
1	0	0	Computer Engineering
2	1	0	Business Administration
3	2	0	Architecture
4	3	1	Mechatronics Engineering
5	4	1	Industrial Design Engineering
6	5	1	Materials Engineering
7	6	2	Computer Engineering
8	7	2	Industrial production
9	8	2	Business Administration

Table students

	ID_STUDENT	ID_CAREER	FIRST_NAME	SECOND_NAME	FIRST_LASTNAME	SECOND_LASTNAME	STUDENT_CARD
1	0	0	Diego	Josue	Mora	Araya	2020456781
2	1	0	Elena	Maria	Diaz	Mora	2021456781
3	2	1	Antonio	Jose	Zambrano	Flores	2022456781
4	3	1	Isabel	Francisca	Torres	Castillo	2023456781
5	4	2	Pilar	Guadalupe	Cruz	Morales	2019456781
6	5	2	Ana	Rosario	Castro	Reyes	2018456781
7	6	3	Marlon	Francisco	Torres	Castillo	2017456781
8	7	3	Pepe	Carlos	Cruz	Morales	2020456744
9	8	4	Joaquin	Fernando	Romero	Moreno	2020556781
10	9	4	Carlos	Alberto	Rivera	Vargas	2020786781
11	10	5	Raquel	Mercedes	Medina	Arias	2020459981
12	11	6	Irene	Teresa	Campos	Sandoval	2020123781
13	12	7	Paula	Petronila	Gallardo	Bravo	2020996781
14	13	8	Beatriz	Julia	Olivares	Orellana	2017098781

Table email

	ID_EMAIL	ID_STUDENT	EMAIL_ADDRESS
1	0	0	josuemora@estudiantec.cr
2	1	1	mariadiaz@estudiantec.cr
3	2	2	josezambranoa@estudiantec.cr
4	3	3	torresfrancisca@estudiantec.cr
5	4	4	guadalupescruz@estudiantec.cr
6	5	5	anarosario@estudiantec.cr
7	6	6	marlonfrancisco@estudiantec.cr
8	7	7	carlospepe@estudiantec.cr
9	8	8	joaquinfernandoa@estudiantec.cr
10	9	9	albertorivera@estudiantec.cr
11	10	10	raquelmercedes@estudiantec.cr
12	11	11	ireneteresa@estudiantec.cr
13	12	12	gallardobravo@estudiantec.cr
14	13	13	juliaolivares@estudiantec.cr

Test:

```
-- =====
--
--                                a. Test getName
-- =====
-- It is executed once to display the prints in the console
SET SERVEROUTPUT ON;

)DECLARE
  vnStudentCard NUMBER(10) := 2020456781;
  vcName VARCHAR2(20);
BEGIN
  vcName := getName(vnStudentCard);
  DBMS_OUTPUT.PUT_LINE('Student name: ' || vcName);

END;

-- .....

)DECLARE
  vnStudentCard NUMBER(10) := 2017098781;
  vcName VARCHAR2(20);
BEGIN
  vcName := getName(vnStudentCard);
  DBMS_OUTPUT.PUT_LINE('Student name: ' || vcName);

END;

-- .....

)DECLARE
  vnStudentCard NUMBER(10) := 2019456781;
  vcName VARCHAR2(20);
BEGIN
  vcName := getName(vnStudentCard);
  DBMS_OUTPUT.PUT_LINE('Student name: ' || vcName);

END;

) -- =====
```

Oracle SQL Developer: C:\Users\david\scripts3\5_Procedures_functions_packages.sql

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 - ID_CAMPUS
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 - ID_STUDENT
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 - GETNAME
 - RETURN
 - PSTUDENTCARD
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 - S_CAREER
 - S_EMAIL
 - S_STUDENT
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 - SE_CAREER
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SQL Worksheet History

Worksheet Query Builder

```
vcName:= getName(vnStudentCard);
DBMS_OUTPUT.PUT_LINE('Student name: ' || vcName);
END;

-----
DECLARE
vnStudentCard NUMBER(10) := 2019456781;
vcName VARCHAR2(20);
BEGIN
vcName:= getName(vnStudentCard);
DBMS_OUTPUT.PUT_LINE('Student name: ' || vcName);
END;

-----
--
-- b. Test getEmail
--
-----
```

Script Output x

Task completed in 0.093 seconds

Student name: Diego

PL/SQL procedure successfully completed.

Student name: Beatriz

PL/SQL procedure successfully completed.

Student name: Pilar

PL/SQL procedure successfully completed.


```

--                                     b. Test getEmail
-- =====

3 DECLARE
  vnIdStudent NUMBER(6) := 0;
  vcEmail VARCHAR2(40);
BEGIN
  vcEmail:= getEmail(vnIdStudent);
  DBMS_OUTPUT.PUT_LINE('Student email: ' || vcEmail);
END;

-- .....

3 DECLARE
  vnIdStudent NUMBER(6) := 13;
  vcEmail VARCHAR2(40);
BEGIN
  vcEmail:= getEmail(vnIdStudent);
  DBMS_OUTPUT.PUT_LINE('Student email: ' || vcEmail);
END;

-- .....

3 DECLARE
  vnIdStudent NUMBER(6) := 4;
  vcEmail VARCHAR2(40);
BEGIN
  vcEmail:= getEmail(vnIdStudent);
  DBMS_OUTPUT.PUT_LINE('Student email: ' || vcEmail);
END;

```

Oracle SQL Developer : C:\Users\david\scripts3\5_Procedures_functions_packages.sql

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Connections

- Tables (Filtered)
 - CAMPUS
 - ID_CAREER
 - ID_CAMPUS
 - CAREER_NAME
 - CAREER
 - ID_CAREER
 - ID_CAMPUS
 - CAREER_NAME
 - EMAIL
 - ID_EMAIL
 - ID_STUDENT
 - EMAIL_ADDRESS
 - STUDENT
 - ID_EMAIL
 - ID_STUDENT
 - EMAIL_ADDRESS
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- Procedures
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 - GETNAME
 - RETURN
 - PSTUDENTCARD
- Operators
- Queues
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- Sequences
 - S_CAMPUS
 - S_CAREER
 - S_EMAIL
 - S_STUDENT
 - S_UNIVERSITY
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 - SE_CAREER
 - SE_EMAIL
 - SE_STUDENT
 - SE_UNIVERSITY
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Compiler - Log x Reports x Welcome Page x 5_Procedures_functions_packages.sql x

SQL Worksheet History

Worksheet Query Builder

```
DECLARE
  vnIdStudent NUMBER(6) := 4;
  vcEmail VARCHAR2(40);
BEGIN
  vcEmail := getEmail(vnIdStudent);
  DBMS_OUTPUT.PUT_LINE('Student email: ' || vcEmail);
END;
```

c. Test getCareer

```
DECLARE
  vnIdStudent NUMBER(6) := 1;
  vcCareer VARCHAR2(50);
BEGIN
  vcCareer := adminCareer.getCareer(vnIdStudent);
```

Script Output x

Task completed in 0.132 seconds

Student email: josuemora@estudiantec.cr

PL/SQL procedure successfully completed.

Student email: juliaolivares@estudiantec.cr

PL/SQL procedure successfully completed.

Student email: guadalupescruz@estudiantec.cr

PL/SQL procedure successfully completed.

```

--                                     c. Test getCareer
-- =====

DECLARE
    vnIdStudent NUMBER(6) := 0;
    vcCareer VARCHAR2(50);
BEGIN
    vcCareer:= adminCareer.getCareer(vnIdStudent);
    DBMS_OUTPUT.PUT_LINE('Student career: ' || vcCareer);
END;

-- .....

DECLARE
    vnIdStudent NUMBER(6) := 13;
    vcCareer VARCHAR2(50);
BEGIN
    vcCareer:= adminCareer.getCareer(vnIdStudent);
    DBMS_OUTPUT.PUT_LINE('Student career: ' || vcCareer);
END;

-- .....

DECLARE
    vnIdStudent NUMBER(6) := 4;
    vcCareer VARCHAR2(50);
BEGIN
    vcCareer:= adminCareer.getCareer(vnIdStudent);
    DBMS_OUTPUT.PUT_LINE('Student career: ' || vcCareer);
END;

```

Oracle SQL Developer : C:\Users\david\scripts3\5_Procedures_functions_packages.sql

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Connections

- Tables (Filtered)
 - CAMPUS
 - CAREER
 - ID_CAREER
 - ID_CAMPUS
 - CAREER_NAME
 - EMAIL
 - ID_EMAIL
 - ID_STUDENT
 - EMAIL_ADDRESS
 - STUDENT
 - UNIVERSITY
- Views
- Indexes
- Packages
- Procedures
- Functions
 - GETEMAIL
 - GETNAME
 - RETURN
 - PSTUDENTCARD
- Operators
- Queues
- Queues Tables
- Triggers
- Types
- Sequences
 - 1,2 S_CAMPUS
 - 1,2 S_CAREER
 - 1,2 S_EMAIL
 - 1,2 S_STUDENT
 - 1,2 S_UNIVERSITY
 - 1,2 SE_CAMPUS
 - 1,2 SE_CAREER
 - 1,2 SE_EMAIL
 - 1,2 SE_STUDENT
 - 1,2 SE_UNIVERSITY
- Materialized Views
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- Database Links
- Public Database Links
- Directories

Compiler - Log Reports Welcome Page 5_Procedures_functions_packages.sql

SQL Worksheet History

Worksheet Query Builder

```
DECLARE
  vnIdStudent NUMBER(6) := 13;
  vcCareer VARCHAR2(50);
BEGIN
  vcCareer:= adminCareer.getCareer(vnIdStudent);
  DBMS_OUTPUT.PUT_LINE('Student career: ' || vcCareer);
END;
```

```
DECLARE
  vnIdStudent NUMBER(6) := 4;
  vcCareer VARCHAR2(50);
BEGIN
  vcCareer:= adminCareer.getCareer(vnIdStudent);
  DBMS_OUTPUT.PUT_LINE('Student career: ' || vcCareer);
END;
```

d. Test getCampus

Script Output x

Task completed in 0.151 seconds

Student career: Computer Engineering

PL/SQL procedure successfully completed.

Student career: Business Administration

PL/SQL procedure successfully completed.

Student career: Architecture

PL/SQL procedure successfully completed.

```

--                                     d. Test getCampus
-- =====

⇒ DECLARE
    vnIdStudent NUMBER(6) := 0;
    vcCampus VARCHAR2(20);
BEGIN
    vcCampus:= adminCampus.getCampus(vnIdStudent);
    DBMS_OUTPUT.PUT_LINE('Student campus: ' || vcCampus);
END;

-- .....

⇒ DECLARE
    vnIdStudent NUMBER(6) := 13;
    vcCampus VARCHAR2(20);
BEGIN
    vcCampus:= adminCampus.getCampus(vnIdStudent);
    DBMS_OUTPUT.PUT_LINE('Student campus: ' || vcCampus);
END;

-- .....

⇒ DECLARE
    vnIdStudent NUMBER(6) := 4;
    vcCampus VARCHAR2(20);
BEGIN
    vcCampus:= adminCampus.getCampus(vnIdStudent);
    DBMS_OUTPUT.PUT_LINE('Student campus: ' || vcCampus);
END;

```

Oracle SQL Developer: C:\Users\david\scripts3\5_Procedures_functions_packages.sql

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Connections

- Tables (Filtered)
 - CAMPUS
 - CAREER
 - ID_CAREER
 - ID_CAMPUS
 - CAREER_NAME
 - EMAIL
 - ID_EMAIL
 - ID_STUDENT
 - EMAIL_ADDRESS
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 - S_CAREER
 - S_EMAIL
 - S_STUDENT
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 - SE_CAMPUS
 - SE_CAREER
 - SE_EMAIL
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 - SE_UNIVERSITY
- Materialized Views
- Materialized View Logs
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- Public Synonyms

Compiler - Log Reports Welcome Page 5_Procedures_functions_packages.sql

SQL Worksheet History

Worksheet Query Builder

```
DECLARE
    vnIdStudent NUMBER(6) := 13;
    vcCampus VARCHAR2(20);
BEGIN
    vcCampus:= adminCampus.getCampus(vnIdStudent);
    DBMS_OUTPUT.PUT_LINE('Student campus: ' || vcCampus);
END;
```

Script Output x

Task completed in 0.07 seconds

Student campus: San Jose

PL/SQL procedure successfully completed.

Student campus: Limon

PL/SQL procedure successfully completed.

Student campus: San Jose

PL/SQL procedure successfully completed.